United States Patent Application Publication	20250283765
Kind Code	A1
Publication Date	September 11, 2025
Inventor(s)	Taylor; Connor et al.

Force Sensitive Sensing

Abstract

An example force-sensitive measuring device includes: a main body having a proximal end and a distal end; a frame coupled to the main body at the proximal end; a spring coupled between the main body and the frame; a sensor coupled to the main body; and a position sensor coupled to the proximal end of the main body, where, when the sensor touches a surface, the spring is compressed and the position sensor detects that the distal end of the main body is touching the surface.

Inventors: **Taylor; Connor (Atlanta, GA), Hake; Sarah (Atlanta, GA)**

Applicant: **Cox Automotive, Inc.** (Atlanta, GA)

Family ID: **96948795**

Appl. No.: **19/075139**

Filed: **March 10, 2025**

Related U.S. Application Data

us-provisional-application US 63562937 20240308

Publication Classification

Int. Cl.: **G01K7/10** (20060101); **G01L1/20** (20060101); **H01M10/42** (20060101); **H01M10/48** (20060101)

U.S. Cl.:

CPC **G01K7/10** (20130101); **G01L1/20** (20130101); **H01M10/4207** (20130101); **H01M10/486** (20130101);

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application claims the benefit of U.S. provisional patent application No. 63/562,937, filed on Mar. 8, 2024, and titled “Force Sensitive Sensing” the disclosure of which is expressly incorporated herein by reference in its entirety.

BACKGROUND

[0002] Sensors are commonly used in both testing and routine operation of various systems. Many types of sensors are configured to sense conditions that are near to, or directly contacting the sensor. Such sensors can be considered “on site” or “in situ.” When sensors are deployed in a system, the sensors can be configured to measure a known location in the system and collect data from that point. The data can then be used to assess whether the system is operating correctly, quantitatively characterize the operation of the system, or for any other purpose.

[0003] There are benefits to improving sensors and devices for using sensors in systems.

SUMMARY

[0004] In some aspects, implementations of the present disclosure relate to a force-sensitive measuring device including: a main body having a proximal end and a distal end; a frame coupled to the main body at the proximal end; a spring coupled between the main body and the frame; a sensor coupled to the main body; and a position sensor coupled to the proximal end of the main body, whereby when the sensor touches a surface, the spring is compressed and the position sensor detects that the distal end of the main body is touching the surface.

[0005] In some aspects, implementations of the present disclosure relate to a force-sensitive measuring device, wherein the sensor is thermocouple and/or thermistor.

[0006] In some aspects, implementations of the present disclosure relate to a force-sensitive measuring device, wherein the spring includes a multi-wave compression spring.

[0007] In some aspects, implementations of the present disclosure relate to a force-sensitive measuring device, wherein the main body includes a channel extending between the proximal end and the distal end, and wherein the sensor is disposed in the channel.

[0008] In some aspects, implementations of the present disclosure relate to a force-sensitive measuring device, wherein the main body includes a bolt with a bolt head formed on the proximal end of the main body.

[0009] In some aspects, implementations of the present disclosure relate to a force-sensitive measuring device, wherein the sensor includes a printed circuit board formed between the bolt head and the frame.

[0010] In some aspects, implementations of the present disclosure relate to a force-sensitive measuring device, wherein the position sensor includes a force-sensitive resistor.

[0011] In some aspects, implementations of the present disclosure relate to a force-sensitive measuring device, wherein the force-sensitive resistor is coupled between the spring and the frame.

[0012] In some aspects, implementations of the present disclosure relate to a force-sensitive measuring device, wherein the main body includes a threaded portion formed on at least a part of a surface extending between the proximal and the distal end.

[0013] In some aspects, implementations of the present disclosure relate to a system for force-sensitive temperature measurement, the system including: an actuator; a frame coupled to the actuator; a controller operably coupled to the actuator, wherein the controller is configured to move the actuator between a deployed position and a retracted position; and a plurality of force-sensitive measuring devices coupled to the frame, wherein each of the plurality of force-sensitive measuring devices includes: a main body having a proximal end and a distal end; a spring coupled between the main body and the frame; a sensor coupled to the main body; and a position sensor coupled to the proximal end of the main body, whereby when the sensor touches a surface, the spring is compressed and the position sensor detects that the distal end of the main body is touching the surface.

[0014] In some aspects, implementations of the present disclosure relate to a system for force-sensitive temperature measurement, wherein the sensor is thermocouple and/or thermistor.

[0015] In some aspects, implementations of the present disclosure relate to a system for force-sensitive temperature measurement, wherein the spring includes a multi-wave compression spring.

[0016] In some aspects, implementations of the present disclosure relate to a system for force-sensitive temperature measurement, wherein the main body includes a channel extending between the proximal end and the distal end, and wherein the sensor is disposed in the channel.

[0017] In some aspects, implementations of the present disclosure relate to a system for force-sensitive temperature measurement, wherein the main body includes a bolt with a bolt head formed on the proximal end of the main body.

[0018] In some aspects, implementations of the present disclosure relate to a system for force-sensitive temperature measurement, wherein the sensor includes a printed circuit board formed between the bolt head and the frame.

[0019] In some aspects, implementations of the present disclosure relate to a system for force-sensitive temperature measurement, wherein the position sensor includes a force-sensitive resistor.

[0020] In some aspects, implementations of the present disclosure relate to a system for force-sensitive temperature measurement, wherein the force-sensitive resistor is coupled between the spring and the frame.

[0021] In some aspects, implementations of the present disclosure relate to a system for force-sensitive temperature measurement, wherein the main body includes a threaded portion formed on at least a part of a surface extending between the proximal and the distal end.

[0022] In some aspects, implementations of the present disclosure relate to a method of making a force-sensitive measuring device, the method including: providing a bolt with a proximal end and a distal end, wherein a bolt head is formed on the proximal end, and a threaded portion including bolt threads extends along a surface of the bolt between the proximal end and the distal end; removing at least a portion of the bolt threads from the bolt to form a smooth section along at least a portion of the surface of the bolt; positioning a printed circuit board adjacent to the head of the bolt, the printed circuit board having an electrical contact; inserting the bolt through a hole formed in a frame; coupling a spring between the bolt and the frame; attaching a nut to the threaded portion of the bolt so that the nut couples the spring, frame, and printed circuit board together; and inserting a sensor through a channel in the bolt towards the distal end of the bolt.

[0023] In some aspects, implementations of the present disclosure relate to a method, wherein the channel in the bolt is formed by boring a channel between the proximal end of the bolt and the distal end of the bolt.

Description

BRIEF DESCRIPTION OF DRAWINGS

[0024] FIG. **1**A illustrates an example force-sensitive measuring device including an electrical contact and spring, where the spring is uncompressed, according to implementations of the present disclosure.

[0025] FIG. **1**B illustrates the example force-sensitive measuring device of FIG. **1**A where the spring is compressed, according to implementations of the present disclosure.

[0026] FIG. **2**A illustrates an example force-sensitive measuring device including a force-sensitive resistor and a spring, where the spring is uncompressed, according to implementations of the present disclosure.

[0027] FIG. **2**B illustrates the example force-sensitive measuring device of FIG. **2**A where the spring is compressed, according to implementations of the present disclosure.
[0028] FIG. **3** illustrates an example battery pack including thermowells, according to implementations of the present disclosure.
[0029] FIG. **4** illustrates an example system including multiple force-sensitive measuring devices, according to implementations of the present disclosure.
[0030] FIG. **5** illustrates a section view of a proximal end of a force-sensitive measuring device with an electrical contact and spring, where the spring is uncompressed, according to implementations of the present disclosure.
[0031] FIG. **6** illustrates an example cross section of a thermowell being measured by an example force-sensitive measuring device, according to implementations of the present disclosure.
[0032] FIG. **7** illustrates an example force-sensitive measuring device, according to implementations of the present disclosure.
[0033] FIG. **8** illustrates a perspective view of the force-sensitive measuring device including a contact sensor, according to implementations of the present disclosure.
[0034] FIG. **9** illustrates an example method of making a force-sensitive measuring device, according to implementations of the present disclosure.
[0035] FIG. **10** illustrates an example computing device.

DETAILED DESCRIPTION

[0036] In-situ sensing can require positioning a sensor so that it is adjacent to, or contacting, a location of interest (e.g., what the sensor is sensing). For example, thermistors and thermocouples are mechanical devices that measure temperature by the change in temperature of the devices themselves. A thermocouple includes two dissimilar metals where a voltage is created across the thermocouple by the Seebeck effect when the dissimilar metals are heated. Likewise, a thermistor experiences a change in resistance when the temperature of the thermistor changes. Thus, thermocouples and thermistors experience the temperature change that they are sensing. Other sensors measuring other physical and chemical phenomena likewise are positioned in situ to sense what the sensor is adjacent to or contacting. As some non-limiting examples, temperature sensors, acid-base sensors, moisture sensors, flow sensors, vibration sensors, and/or chemical sensors, all can require that the sensor be disposed close to the subject that is sensed by the sensor. While various sensors can benefit from being placed in-situ, placing sensors in situ can risk damage to the sensor or to an object being sensed. The present disclosure includes systems, devices, and methods for force-sensitive measurement, where the example systems, devices, and methods can position sensors to contact or be adjacent to a subject (e.g., the surface of the subject), while limiting the force applied to a location being sensed (e.g., to prevent damage).
[0037] With reference to FIGS. **1**A and **1**B, implementations of the present disclosure include a force-sensitive measuring device **100**. The force-sensitive measuring device **100** includes a main body **110**. The main body **110** includes a proximal end **112** and a distal end **114**.
[0038] Optionally, the proximal end **112** of the main body **110** can be configured to attach to a frame **120**. The frame **120** can be configured to attach to any number of force-sensitive measuring devices **100**. As shown in FIGS. **1**A and **1**B, the main body **110** can optionally include a bushing **122** between the main body **110** and the frame **120**.
[0039] Still with reference to FIGS. **1**A and **1**B, the force-sensitive measuring device **100** can include a spring **130**. The spring **130** can be disposed between the frame **120** and the main body **110**, so that a force applied to the main body **110** can cause the spring **130** to be compressed or extended from the resting length of the spring **130**. As shown in FIGS. **1**A and **1**B, the bushing **122** can be used to attach the spring **130** to the main body **110**. Optionally, the spring **130** can be a helical wave spring including flat wire windings, but implementations of the present disclosure can include any type of spring, including coil springs, multi-wave compression springs, leaf springs, and other wave springs. In some implementations the spring **130** can be a section of elastic material

or compressible material.

[0040] In some implementations of the present disclosure, the main body **110** can be formed from a bolt and/or using a bolt. For example, the proximal end **112** of the main body **110** can be shaped as a bolt head, and a threaded portion **118** can be disposed over any portion of the main body **110** between the proximal end **112** and distal end **114** of the main body **110**. Optionally, a nut **124** can be attached to the threaded portion **118** and configured to couple a spring **130** between the frame **120** and the main body **110**, as shown in FIG. **1**A and FIG. **1**B.

[0041] The force-sensitive measurement device **100** can include a position sensor **140** that detects the movement of the main body **110** relative to the frame **120**. For example, a force can be applied to the distal end **114** of the main body toward the frame **120**, causing the spring **130** to be compressed. The position sensor **140** can optionally be configured to measure the amount of compression of the spring **130**, movement of the spring **130**, or a force exerted by the spring **130**.

[0042] As used herein, a position sensor can refer to sensors that detect or measure force or movement of the main body **110**. The position sensor **140** can be any type of force or contact sensor, including various types of load cells (pneumatic, piezoelectric capacitive, inductive, magnetostrictive), strain gauge(s), electrical contacts, force-sensitive resistors etc. In the implementation of a force-sensitive measurement device **100** of FIGS. **1**A and **1**B, the position sensor **140** is an electrical contact. The position sensor **140** can further include a Printed Circuit Board (PCB) **142**. Optionally, the PCB **142** can be a square PCB that is configured to be disposed in the frame **120**, as shown in FIGS. **1**A and **1**B. If the proximal end **112** of the main body **110** is shaped as a bolt head, the PCB **142** can be configured to be attached to the main body **110** by the bolt head and bushing **122**. The position sensor **140** can form a closed circuit to the frame **120** when the spring **130** is in an extended and/or relaxed state, as shown in FIG. **1**B. When a force is applied compressing the spring **130**, the position sensor and PCB are lifted off the frame **120** as shown in FIG. **1**A. Lifting the position sensor **140** off the frame **120** creates an open circuit with the contact of the position sensor **140**, indicating that force is being applied to the main body **110**.

[0043] The distal end **114** of the main body **110** can include a sensor **150**. The main body **110** can be sized and shaped to position the sensor **150** adjacent to or contacting an area that the sensor **150** is configured to sense. In some implementations, the sensor **150** is a thermocouple and/or thermistor configured to measure the temperature of a surface adjacent to the distal end **114** of the main body **110**.

[0044] Optionally, the main body **110** can be formed from a material selected to improve the performance of the sensor **150**. For example, if the sensor **150** is a temperature sensor, the main body can be formed from a temperature insulating material that limits the absorption of heat by the main body **110** to improve the accuracy of temperature measurement from the sensor **150**. In some implementations, the main body **110** is formed from a synthetic polymer (e.g., a nylon bolt or shaft) that insulates the sensor **150** to improve the accuracy of the temperature reading.

[0045] Optionally, as shown in FIG. **1**A and FIG. **1**B, the main body **110** can include a channel **116** formed between the proximal end **112** and distal end **114** of the main body **110**. The channel can be configured to receive the sensor **150** and/or any electrical connections with the sensor **150** (e.g., wires).

[0046] In some implementations of the present disclosure, the position sensor **140** can be used to determine when the sensor **150** contacts a surface being measured, and/or to prevent the main body and/or sensor **150** from applying excessive force to the surface being measured. For example, the spring **130** can be sized so that the amount of force required to cause a movement of the main body **110** relative to the frame **120** is small enough that a surface or subject being sensed is not damaged. Implementations of the force-sensitive measuring device described herein can therefore provide a sensor that can be deployed to different lengths depending on where the surface being sensed is relative to the frame. Optionally, this can benefit sensing systems where the tolerances and configurations of parts are such that the exact location of a surface being sensed is not certain.

[0047] With reference to FIGS. **2**A and **2**B, implementations of the present disclosure include force-sensitive measuring devices **200** including a force-sensitive resistor **240** as a position sensor. The force-sensitive measuring device shown in FIGS. **2**A and **2**B includes the frame **120**, spring **130**, bushing **122**, nut **124**, main body **110**, channel **116**, and sensor **150** shown and described in FIGS. **1**A and **1**B. The force-sensitive resistor **240** can be disposed between the spring **130** and the frame **120** so that a force applied to the distal end **114** of the main body **110** is transmitted through the spring **130** and to the force-sensitive resistor **240**. By measuring the resistance of the force-sensitive resistor **240**, the amount of force on the main body **110** can be measured/detected, allowing a detection that the force-sensitive measuring device **200** is contacting a surface with the sensor **150** located at the distal end **114** of the main body **110**.

[0048] FIG. **2**A illustrates the force sensitive measuring device **200** where a force is not applied to the distal end **114** in the direction of the proximal end **112**. The spring **130** is in a relaxed state. In the relaxed state, the spring **130** can apply some force to the force-sensitive resistor **240**. FIG. **2**B illustrates the force sensitive measuring device **200** when a force is applied to the distal end **114** in the direction of the proximal end **112**. As shown in FIG. **2**B, the spring **130** compresses in response to force being applied to the distal end **114** towards the proximal end **112**, increasing the force on the force-sensitive resistor **240** relative to the force on the force sensitive resistor **240** when the spring **130** is in a relaxed state.

[0049] With reference to FIG. **3**, an example battery pack **300** is illustrated that can be measured using the force-sensitive measuring devices **100**, **200** shown in FIGS. **1**A-**2**B. The battery pack **300** includes one or more battery modules **302**. A row of thermowells **304** runs along the battery pack, where each battery includes one thermowell. As used herein, a “thermowell” refers to a structure that is configured to allow a probe to measure a temperature from a location near the center of the battery (or other subject), without the probe being located inside the battery module (or other subject). As described with reference to FIGS. **1**A-**2**B, the main body **110** of the force-sensitive measuring device **100**, **200** can optionally include a temperature sensor (e.g., a thermocouple or thermistor). The main body **110** of one or more force-sensitive measuring devices **100**, **200** can be inserted into the thermowells **304** of the one or more battery modules **302** to measure the temperature of the battery modules **302**. Because the force-sensitive measuring devices **100**, **200** include springs that allow the main body to move relative to the frame, the force-sensitive measuring device **100**, **200** can detect when a surface of the thermowells is touched and also prevent excessive force from being applied to the thermowell. This allows the sensor **150** to be positioned accurately inside the battery module **302** and/avoids damaging the battery module **302**. It should be understood that the battery pack **300** is a non-limiting example of a structure that can be measured using implementations of the present disclosure, and that various implementations of the present disclosure can be used for any application, and can include any type of sensors, as described with reference to FIGS. **1**A-**2**B.

[0050] With reference to FIG. **4**, implementations of the present disclosure can include systems for force-sensitive measurement **400**, including an array **410** of force-sensitive measuring devices (e.g., the force-sensitive measuring devices **100**, **200** described with reference to FIGS. **1**A-**2**B) attached to a common frame **420**. The array **410** and frame **420** can be sized so that the array **410** fits within a set of thermowells (e.g., the thermowells **304**, shown in FIG. **3**).

[0051] Optionally, implementations of the present disclosure can include an actuator (not shown) configured to move the array **410**, and/or a controller configured to move the actuator. Optionally, the controller can be configured to receive measurements from the position sensors and/or sensors of the one or more force-sensitive measuring devices. Optionally, the movement of the actuator is based on measurements from one or more positions sensors of the force-sensitive measuring devices. For example, the controller can be configured to cause the actuator to lower the array **410** into thermowells, and stop lowering the array when a condition is met (e.g., one or more position sensors of the force-sensitive measuring devices detects contact with a surface). Optionally, the controller can include any or all of the components of the computing device **1000** shown in FIG. **10**.

[0052] With reference to FIG. **5**, a sectional view is shown of a proximal end **112** of an example implementation of a force-sensitive measuring device, such as the measuring devices **100**, **200** of FIG. **1**A-**2**B. The sectional view illustrates a coupling between the frame **120**, PCB **142**, position sensor **140**, spring **130**, bushing **122**, and nut **124**, described with reference to FIGS. **1**A-**1**B.

[0053] With reference to FIG. **6**, implementations of the present disclosure can optionally be configured for force-sensitive measurements without using the position sensor **140** or force-sensitive resistor **240** described with reference to FIGS. **1**A-**2**B. FIG. **6** illustrates a thermowell **610** that the sensor **150** is configured to measure the temperature of. The spring **130** prevents the distal end **114** of the main body from applying excessive force to the thermowell **610**, and thereby can prevent damage to the thermowell **610**, while also ensuring that the sensor **150** contacts the thermowell **610**.

[0054] FIG. **7** illustrates the example implementation shown in FIG. **6** in an uncompressed position **710** and a compressed position **720**.

[0055] FIG. **8** illustrates a perspective view of an example implementation of the present disclosure including a position sensor **140** positioned between the spring **130** and frame **120**.

[0056] With reference to FIG. **9**, implementations of the present disclosure include methods of making force-sensitive measuring devices.

[0057] At step **910**, the method includes providing a bolt with a proximal end and a distal end, where a bolt head is formed on the proximal end, and a threaded portion comprising bolt threads extends along a surface of the bolt between the proximal end and the distal end. FIGS. **1**A-**2**B illustrate an example main body **110** that can optionally be formed as a bolt.

[0058] At step **920** the method includes removing at least a portion of the bolt threads from the bolt to form a smooth section along at least a portion of the surface of the bolt. For example, the bolt may include threads along the whole surface between the proximal and distal end of the bolt, and the method can include removing unnecessary threads toward the distal end of the bolt. As shown in FIG. **1**A-**2**B, the main body **110** includes a threaded portion **118** that terminates before the distal end **114** of the main body **110**.

[0059] At step **930**, the method includes positioning a printed circuit board adjacent to the head of the bolt, the printed circuit board having an electrical contact. FIGS. **1**A-**1**B illustrate an example PCB **142** and position sensor **140** with an electrical contact.

[0060] At step **940**, the method includes inserting the bolt through a hole formed in a frame. As shown in the cross-sectional view in FIGS. **1**A-**2**B, the main body **110** passes through the frame **120**.

[0061] At step **950**, the method includes coupling a spring between the bolt and the frame. As described with reference to FIGS. **1**A-**2**B, the spring can be any kind of spring, and can optionally be coupled between the frame and the main body using a bushing **122** as shown in FIGS. **1**A-**2**B.

[0062] At step **960**, the method includes attaching a nut to the threaded portion of the bolt so that the nut couples the spring, frame, and printed circuit board together. FIGS. **1**A-**1**B illustrate a nut **124** that couples the PCB **142** to the main body **110**.

[0063] At step **970**, the method includes inserting a sensor through a channel in the bolt towards the distal end of the bolt. As shown in FIGS. **1**A-**2**B, the main body **110** can include a channel between the proximal end **112** and distal end **114**. In some implementations, the method can further include boring a channel between the proximal end of the bolt and the distal end of the bolt.

[0064] Referring to FIG. **10**, an example computing device **1000** upon which the methods described herein may be implemented is illustrated. It should be understood that the example computing device **1000** is only one example of a suitable computing environment upon which the methods described herein may be implemented. Optionally, the computing device **1000** can be a well-known computing system including, but not limited to, personal computers, servers, handheld or laptop devices, multiprocessor systems, microprocessor-based systems, network personal computers (PCs), minicomputers, mainframe computers, embedded systems, and/or distributed computing environments including a plurality of any of the above systems or devices. Distributed computing environments enable remote computing devices, which are connected to a communication network or other data transmission medium, to perform various tasks. In the distributed computing environment, the program modules, applications, and other data may be stored on local and/or remote computer storage media.

[0065] In its most basic configuration, computing device **1000** typically includes at least one processing unit **1006** and system memory **1004**. Depending on the exact configuration and type of computing device, system memory **1004** may be volatile (such as random access memory (RAM)), non-volatile (such as read-only memory (ROM), flash memory, etc.), or some combination of the two. This most basic configuration is illustrated in FIG. **3** by dashed line **1002**. The processing unit **1006** may be a standard programmable processor that performs arithmetic and logic operations necessary for operation of the computing device **1000**. The computing device **1000** may also include a bus or other communication mechanism for communicating information among various components of the computing device **1000**.

[0066] Computing device **1000** may have additional features/functionality. For example, computing device **1000** may include additional storage such as removable storage **1008** and non-removable storage **1010** including, but not limited to, magnetic or optical disks or tapes. Computing device **1000** may also contain network connection(s) **1016** that allow the device to communicate with other devices. Computing device **1000** may also have input device(s) **1014** such as a keyboard, mouse, touch screen, etc. Output device(s) **1012** such as a display, speakers, printer, etc. may also be included. The additional devices may be connected to the bus in order to facilitate communication of data among the components of the computing device **1000**. All these devices are well known in the art and need not be discussed at length here.

[0067] The processing unit **1006** may be configured to execute program code encoded in tangible, computer-readable media. Tangible, computer-readable media refers to any media that is capable of providing data that causes the computing device **1000** (i.e., a machine) to operate in a particular fashion. Various computer-readable media may be utilized to provide instructions to the processing unit **1006** for execution. Example tangible, computer-readable media may include, but is not limited to, volatile media, non-volatile media, removable media and non-removable media implemented in any method or technology for storage of information such as computer readable instructions, data structures, program modules or other data. System memory **1004**, removable storage **1008**, and non-removable storage **1010** are all examples of tangible, computer storage media. Example tangible, computer-readable recording media include, but are not limited to, an integrated circuit (e.g., field-programmable gate array or application-specific IC), a hard disk, an optical disk, a magneto-optical disk, a floppy disk, a magnetic tape, a holographic storage medium, a solid-state device, RAM, ROM, electrically erasable program read-only memory (EEPROM), flash memory or other memory technology, CD-ROM, digital versatile disks (DVD) or other optical storage, magnetic cassettes, magnetic tape, magnetic disk storage or other magnetic storage devices.

[0068] In an example implementation, the processing unit **1006** may execute program code stored in the system memory **1004**. For example, the bus may carry data to the system memory **1004**, from which the processing unit **1006** receives and executes instructions. The data received by the system memory **1004** may optionally be stored on the removable storage **1008** or the non-removable storage **1010** before or after execution by the processing unit **1006**.

[0069] It should be understood that the various techniques described herein may be implemented in connection with hardware or software or, where appropriate, with a combination thereof. Thus, the methods and apparatuses of the presently disclosed subject matter, or certain aspects or portions thereof, may take the form of program code (i.e., instructions) embodied in tangible media, such as floppy diskettes, CD-ROMs, hard drives, or any other machine-readable storage medium wherein, when the program code is loaded into and executed by a machine, such as a computing device, the machine becomes an apparatus for practicing the presently disclosed subject matter. In the case of program code execution on programmable computers, the computing device generally includes a processor, a storage medium readable by the processor (including volatile and non-volatile memory and/or storage elements), at least one input device, and at least one output device. One or more programs may implement or utilize the processes described in connection with the presently disclosed subject matter, e.g., through the use of an application programming interface (API), reusable controls, or the like. Such programs may be implemented in a high level procedural or object-oriented programming language to communicate with a computer system. However, the program(s) can be implemented in assembly or machine language, if desired. In any case, the language may be a compiled or interpreted language and it may be combined with hardware implementations.

Claims

1. A force-sensitive measuring device comprising: a main body having a proximal end and a distal end; a frame coupled to the main body at the proximal end; a spring coupled between the main body and the frame; a sensor coupled to the main body; and a position sensor coupled to the proximal end of the main body, whereby when the sensor touches a surface, the spring is compressed and the position sensor detects that the distal end of the main body is touching the surface.
2. The force-sensitive measuring device of claim 1, wherein the sensor is thermocouple.
3. The force-sensitive measuring device of claim 1, wherein the spring comprises a multi-wave compression spring.
4. The force-sensitive measuring device of claim 1, wherein the main body comprises a channel extending between the proximal end and the distal end, and wherein the sensor is disposed in the channel.
5. The force-sensitive measuring device of claim 1, wherein the main body comprises a bolt with a bolt head formed on the proximal end of the main body.
6. The force-sensitive measuring device of claim 5, wherein the sensor comprises a printed circuit board formed between the bolt head and the frame.
7. The force-sensitive measuring device of claim 1, wherein the position sensor comprises a force-sensitive resistor.
8. The force-sensitive measuring device of claim 7, wherein the force-sensitive resistor is coupled between the spring and the frame.
9. The force-sensitive measuring device of claim 1, wherein the main body comprises a threaded portion formed on at least a part of a surface extending between the proximal and the distal end.
10. A system for force-sensitive temperature measurement, the system comprising: an actuator; a frame coupled to the actuator; a controller operably coupled to the actuator, wherein the controller is configured to move the actuator between a deployed position and a retracted position; and a plurality of force-sensitive measuring devices coupled to the frame, wherein each of the plurality of force-sensitive measuring devices comprises: a main body having a proximal end and a distal end; a spring coupled between the main body and the frame; a sensor coupled to the main body; and a position sensor coupled to the proximal end of the main body, whereby when the sensor touches a surface, the spring is compressed and the position sensor detects that the distal end of the main body is touching the surface.
11. The system for force-sensitive temperature measurement of claim 10, wherein the sensor is a thermocouple.
12. The system for force-sensitive temperature measurement of claim 10, wherein the spring

comprises a multi-wave compression spring.

13. The system for force-sensitive temperature measurement of claim 10, wherein the main body comprises a channel extending between the proximal end and the distal end, and wherein the sensor is disposed in the channel.

14. The system for force-sensitive temperature measurement of claim 10, wherein the main body comprises a bolt with a bolt head formed on the proximal end of the main body.

15. The system for force-sensitive temperature measurement of claim 14, wherein the sensor comprises a printed circuit board formed between the bolt head and the frame.

16. The system for force-sensitive temperature measurement of claim 10, wherein the position sensor comprises a force-sensitive resistor.

17. The system for force-sensitive temperature measurement of claim 16, wherein the force-sensitive resistor is coupled between the spring and the frame.

18. The system for force-sensitive temperature measurement of claim 10, wherein the main body comprises a threaded portion formed on at least a part of a surface extending between the proximal and the distal end.

19. A method of making a force-sensitive measuring device, the method comprising: providing a bolt with a proximal end and a distal end, wherein a bolt head is formed on the proximal end, and a threaded portion comprising bolt threads extends along a surface of the bolt between the proximal end and the distal end; removing at least a portion of the bolt threads from the bolt to form a smooth section along at least a portion of the surface of the bolt; positioning a printed circuit board adjacent to the head of the bolt, the printed circuit board having an electrical contact; inserting the bolt through a hole formed in a frame; coupling a spring between the bolt and the frame; attaching a nut to the threaded portion of the bolt so that the nut couples the spring, frame, and printed circuit board together; and inserting a sensor through a channel in the bolt towards the distal end of the bolt.

20. The method of claim 19, wherein the channel in the bolt is formed by boring a channel between the proximal end of the bolt and the distal end of the bolt.